

Rebecca Steele Wanger

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EDUCATION

Ph.D., Forestry, Ecology, Evolution, and Behavior Michigan State University **June 2031**

Advisor: Dr. Andrew Finley

Bachelor of Arts Colby College, **May 2025**

Major: Environmental Science, Biology: Ecology and Evolution **GPA: 3.99** *summa cum laude*

Minor: Statistics

Honors Thesis: Identifying the Spread of Invasive Jumping Worms across Maine's Forests using Soil eDNA

Study Abroad Round River Conservation Studies, Chilean Patagonia, **Fall 2023**

RESEARCH INTERESTS

I am interested in the statistical modeling and analysis of forest and climate change ecology, conservation, pathology, ecophysiology, species interactions, and genetics and genomics.

GRANTS, AWARDS, AND SPONSORED RESEARCH

EEB Engagement Fellowship, Michigan State University 2026

Honors Distinction in Biology, Colby College 2025

Honors Distinction in Environmental Science, Colby College 2025

F. Russell Cole Student Research Fellowship, Colby College 2024-2025

Dana Scholar, Colby College 2022-2023, 2024-2025

Mu Sigma Rho, Colby College, National Statistics Honor Society 2024-2025

Research Experience Extension Fellow, The Morton Arboretum 2025

Research Experiences for Undergraduates, The Morton Arboretum, NSF 2024

Dean's List, Colby College All Attending Semesters 2021-2025

Phi Beta Kappa, Colby College, Junior Inductee 2024

RESEARCH EXPERIENCE

Productivity-Resilience Trade-Offs in Southern Pine: Coordinated Stress Response Mechanisms to Drought and Brown Spot Needle Blight

Research Technician, University of Georgia

January 2026 – June 2026

Advisor: Dr. Helen Bothwell

- Planned and underwent the logistics for a greenhouse trial, subjecting loblolly and longleaf pine to drought and *Lecanosticta acicola* (pathogen agent causing brown spot needle blight)
- Selected, cold stratified, sowed, germinated, and transplanted thousands of loblolly and longleaf pine in growth chambers and greenhouses
- Managed and updated a detailed project guide and data on seedling growth and health
- Analyzed preliminary results of mortalities and effective growth strategies
- Isolated and cultured *Lecanosticta acicola* for inoculation

Identifying the Spread of Invasive Jumping Worms in Maine's Forests

Environmental Science Honors Thesis, Colby College

February 2024 – May 2025

Advisors: Dr. Justin Becknell and Dr. Pete Countway

- Worked to track the spread of invasive worms across the state's protected forests and examine the relationship between the worms' presence and plant/soil composition
- Independently designed the protocols and data sheets and collected the site, plant composition, soil chemistry, and soil eDNA data from 57 177 m² circular forested plots in Maine's public reserved lands, state parks, botanical gardens, arboretums, and land trusts
- Extracted, amplified, purified, and sequenced COI regions of six earthworm tissue samples and 57 soil eDNA samples using PCR, Agar Gel Electrophoresis, and Oxford Nanopore Sequencing
- Cleaned and mapped sequenced reads against a database of 4,013 reference sequences of jumping (*Amyntas* spp.) and European worms (*Lumbricus* spp.) using Geneious
- Performed extensive literature search culminating in writing a literature review on what is known about Jumping Worms in northeastern forest ecosystems
- Wrote and presented honor's thesis research (136 pg.) and analysis, including maps of sample plots and findings made in ArcGIS

Microbial Source Tracking of *Campylobacter jejuni* in the Water and Soil of Casco Bay, ME

Dr. Pete Countway and Dr. David Ernst, Colby College **September 2024 – December 2024**

- Used qPCR to detect the presence and concentration of *C. jejuni* in water and sediment samples
- Extracted and quantified DNA from filtered water and sediment samples using the DNeasy PowerSoil Pro Kit and Qubit Flex Fluorometer
- Collected and filtered water, sediment, and nutrient samples using sterile technique and microbial filtering rig
- Wrote a literature review and literature journal using over 25 primary sources on the epidemiology of *C. jejuni* and molecular methods for detecting and source tracking them
- Wrote a 30-page scientific paper and helped write and present an 85-slide presentation summarizing and discussing our results

Analyzing the spatial, morphological, and environmental characteristics of *Quercus brandegeei*

Dr. Sean Hoban, The Morton Arboretum, REU **May 2024 – Aug. 2024, Jan. 2025 – July 2025**

- Analyzed correlations between morphological, environmental, and spatial traits of an endemic and endangered tree species of Baja California Sur, Mexico
- Formulated and examined hypotheses with appropriate statistics in ArcGIS and R Studio, using base R, Tidyverse, sf, raster and other packages to run Ripley's K, ANN analysis, Point Poisson Models, simple and multiple linear regression, and GAMs, [GitHub Link](#)
- Extensively documented all methodologies performed for future scientific publication
- Extracted and quantified DNA from 394 leaf tissue samples of *Q. brandegeei* individuals and assisted with microsatellite analysis in Geneious
- Actively participated in weekly career and professional development seminars and journal clubs
- Assisted with fieldwork involving morphological data and tissue collection from trees

Four Years of Change in Maine's Forests

Dr. Justin Becknell, Colby College, Summer Research Assistant

May 2022 – August 2022

- Collected forest inventory data by surveying species abundance, richness, canopy position, and crown illumination
- Analyzed forest inventory data to track changes in above-ground biomass, tree species abundance, and tree mortality across time
- Designed and presented a poster summarizing this research project
- Designed a novel protocol for Song Meter Micro, a wildlife acoustic recording device

Applications of Terrestrial Laser Scanning in Forest Ecology

Dr. Justin Becknell, Colby College, Summer Research Assistant

May 2022 – August 2022

- Collected RIEGL VZ-400i Terrestrial Laser Scanner point cloud scans of forested circular plots for extracting plant morphological and structural data and expediting forest inventories
- Experimented with optimal locations of the laser, scan reflectors, and the scanner angles
- Designed a novel protocol for the RIEGL VZ-400i Terrestrial Laser Scanner

PUBLICATIONS

Wanger, R. "Identifying the Spread of Invasive Jumping Worms across Maine's Forests using Soil eDNA" (2025). Colby College Digital Commons. *Honors Theses*. Paper 1498.

PRESENTATIONS

Wanger, R., Hamilton, A., Perez Morales, D. W., Alvarez-Clare, S., and Hoban, S. "Exploring the demographic, spatial, and environmental traits of the climate relict Arroyo Oak to investigate water dependency." Botany Conference, Palm Springs, CA, July 2025. Oral Presentation.

Wanger, R. "Identifying the Spread of Invasive Jumping Worms across Maine's Forests using Soil eDNA." Colby Liberal Arts Symposium, Colby College, May 2025. Oral Presentation.

Wanger, R. "Identifying the Spread of Invasive Jumping Worms across Maine's Forests using eDNA." Northeast Natural History Conference, Springfield, MA, April 2025. Oral Presentation.

Wanger, R. "The Spread of Invasive Jumping Worms in Maine." Environmental Studies Summer Research Presentations, Colby College, September 2024. Oral Presentation.

Wanger, R. "Exploring the morphological, spatial, and environmental traits of the endangered Arroyo Oak." Center for Tree Science Undergraduate Research Symposium, The Morton Arboretum, July 2024. Oral presentation.

Wanger, R. "Exploring the relationships between the environmental, morphological, spatial, and genetic traits of the Arroyo Oak to help inform Conservation." Elmhurst Scholars Research Summit, Elmhurst University, July 2024. Oral presentation – Lightning talk.

Mohan, R., Amin, S., and **Wanger, R.** "Applications of Terrestrial Laser Scanning in Forest Ecology." Colby Undergraduate Summer Research Retreat (CUSRR), Colby College, August 2022. Poster.

Wanger, R. “Four Years of Change in Maine Forests.” Colby Undergraduate Summer Research Retreat (CUSRR), Colby College, August 2022. Poster.

RELATED PROFESSIONAL EXPERIENCES

Biological Science Technician Plants, NPS, **June 25’ – Nov. 25’, June 26’ – August 26’**
Sleeping Bear Dunes National Lake Shore, Empire, Michigan

- Surveyed for Oak Wilt Disease (OWD) and hemlock woolly adelgid (HWA) in high probability areas and for remote sensing ground truthing
- Treated around 90 white ash trees affected by EAB with tree injections using pesticides
- Surveyed for rare plants, such as Pitcher’s Thistle (*Cirsium pitcheri*), Green Comet Milkweed (*Asclepias viridis*), and Blue Flag Iris (*Iris virginica*)
- Tested for OWD using branch samples and the AgroLAMP Oak Wilt Detection Kit
- Treated over 100 eastern hemlock trees with tree injections using pesticides and delimited over 200 for HWA prevention
- Mechanically and chemically treated baby’s breath (*Gypsophila paniculata*) and slender false brome (*Brachypodium sylvaticum*)
- Set up resistance tests for Beech Bark Disease by applying pads with beech scale

Plant Care Technician, Colby College, **Sept. 2022 – May 2025**
Waterville, Maine

- Watered, monitored, repotted, and propagated plants in Colby’s greenhouse four times a week
- Grew plants from seed and via propagation
- Treated plants for scale, mealy bugs, fungus gnats, spider mites, and thrip with pesticides
- Helped organize and run a plant sale to Colby students and faculty

Field Assistant, Round River Conservation Studies **Sept. 2023 – Dec. 2023**
Claudia Scognamillo, Patagonia, Chile

- Led trainings for 20 people on a protocol I wrote for a rapid biodiversity inventory (RBI) assessment
- Led two teams of five on two different backpacking trips (two and three days long) to collect RBI data
- Censused Guanaco populations with rangefinders, binoculars, and compasses for three days along line transects
- Collected growth data on reforestation plots of *Nothofagus antartica* with a team of ten
- Surveyed Magellanic woodpeckers and their cavities for three days along line transects

Field Ecology Assistant, Maine Natural Areas Program, **June 2023 – August 2023**
Justin Schlawin, Augusta, Maine

- Inventoried 75 Ecological Reserve plots using the MNAP Ecological Reserve protocol for the Bureau of Parks and Lands
- Entered data and organized field forms and gear from the EcoReserve plots
- Wrote and created maps in ArcGIS for a Natural Resource Inventory on the Androscoggin Riverlands State Park
- Wrote and filed site survey summaries and rare plant forms
- Created maps in ArcGIS for field days

Student Math Grader Calculus II, Colby College,
Dr. Nora Youngs, Waterville, Maine

Feb. 2022 – May 2022

- Graded homeworks and labs for a single-variable calculus class of 30 students
- Suggested ways to improve teaching based on observed learning gaps

Environmental Studies Research Assistant, Colby College,
Dr. Justin Becknell, Waterville, Maine

Nov. 2021 – May 2022

- Cleaned and organized the lab
- Distilled a complex workflow into a succinct protocol for an automated soil particle size analyzer for student workers

Crew Member, Student Conservation Association,
Daiva Gyls, Chicago, Illinois

July 2020 – August 2020

- Painted a mural, planted native species, and managed invasive plants with loppers and hand saws

LEADERSHIP EXPERIENCES

Team Member/Captain, *Colby Woodsmen Team*, Waterville, Maine **Sept. 2021 – May 2025**

- Captained the team in May 2024 to May 2025, organizing and running team practices, communications, alumni outreach, funding opportunities, meets, and social gatherings
- Competed in over 20 intercollegiate timber sport competitions in teams of 6, wielding axes, peavies, hatchets, bow saws, and two-man crosscut saws

First-Year Experience Leader, *Colby College*, Waterville, Maine

August 2024

- Led ten first year students on an outdoor orientation basecamp trip for three days
- Actively participated in a week of training sessions on leading student groups outdoors

Percussionist, *Colby Wind Ensemble*, Waterville, Maine

Sept. 2021 – May 2025

- Participated in Wind Ensemble rehearsals (2 hrs/week) and seven concerts

Mentor, *Colby Cares about Kids*, Waterville, Maine

Sept. 2021 – May 2024

- Participated in workshops about mentoring kids in Waterville
- Mentored a middle school girl for an hour a week

ES Student Board Member, Colby College, Waterville, Maine

Sept. 2022 – May 2023

- Planned, organized, and ran three departmental social events for over 100 people
- Spoke with prospective professors about Colby and the department

SKILLS

Computer and Data Analysis

- Proficient with R for running statistical analyses, cleaning data sets, and creating graphs with tidyverse, spatstat, sf, raster, and base R.
- Proficient with ArcGIS for data manipulation, mapping, and different spatial analysis statistics like ANN analysis, Poisson Point Models, Spatial Autocorrelation, and Interpolation
- Proficient with Google Sheets and Excel for managing and cleaning large data sets
- Experience with using Python for statistical analysis and managing data

- Experienced with using Geneious to trim, map, blast, align, analyze microsatellites, and form trees with sequences
- Experienced with using Linux for comparative genomic analysis

Field Work

- Experienced with the collection and processing of plant leaf tissue, water, and soil samples using sterile techniques for DNA and nutrient data
- Experienced with identifying trees and vegetation in northeastern forests
- Experienced with fieldwork and forestry tools, such as rangefinders, GPS devices, increment bores, clinometers, DBH tapes, and prisms
- Experienced using electronic (ESRI Field Maps, Survey123) and paper surveys
- Experienced with stream macroinvertebrate collection and identification
- Experienced surveying for forest pests, such as Oak Wilt Disease, Emerald Ash Borers, Beech Bark Disease, Beech Leaf Disease, and Hemlock Woolly Adelgid
- Experienced with using tree injector and hand-held sprayers for applying pesticides and herbicides for tree pest and pathogen treatments

Lab Work

- Experienced with microbiological techniques like serial dilutions, microbial isolations from soil samples, staining, culturing, and sterile technique
- Experienced with dilutions, centrifuges, vortexing, water baths, and pipetting
- Proficient in plant DNA extraction and quantification (Nanodrop and Qubit)
- Experienced with Oxford Nanopore Sequencing
- Experienced with Agar Gel Electrophoresis and gel staining
- Experienced with PCR and qPCR primer and probe design and methodologies
- Experienced with water quality measurements, using probes (conductivity, discharge, temperature, etc.) and nutrient and chlorophyll analysis

Languages

- Native English speaker

TRAINING AND CERTIFICATIONS

Non-Wildland Fire Chainsaw Operator Sawyer Certificate of Completion, **November 2025**
National Park Service

Certified Pesticide Applicator in Michigan Certificate of Completion, **November 2025**
Commercial (Core, 2, 3B, 6)
MDARD

Adult First AID CPR AED Certificate of Completion, **August 2025**
Health & Safety Institute
Certificate link:
<https://otis.osmanager4.com/#!/validation?first=Rebecca&last=Wanger&code=C649319B5>

Wilderness First Aid, Certificate of Completion, **May 2024**
Wilderness Medical Associates International
Certificate Link:
<https://courses.wildmed.com/mod/customcert/view.php?id=122870&downloadown=1>

MITx, edX

Certificate of Completion, **August 2022**

6.00.1x: Introduction to Computer Science and Programming Using Python

RELATED COURSEWORK

- **Plant Science**
 - Plant Physiology, Genetics, Genomics, Cellular Biology, Earth Systems Chemistry, Ecology, Evolution and Diversity, Evolutionary Analysis, Forest Ecosystems, Agroecosystems, Microbiology, Aquatic Ecosystems, Plant Evolution, Environment and Society, Ecology in Maine
- **Data Analysis**
 - GIS and Spatial Analysis, Statistical Modeling, Epidemiology, Statistical Learning in Data Science, Population Modeling, Intro to Statistics and Data Science, Statistical Graphics, Linear Algebra, Series and Multivariable Calculus

REFERENCES

Dr. Sean Hoban
Tree Conservation Biologist
Hoban Lab
The Morton Arboretum
(630)-719-2419
shoban@mortonarb.org

Dr. Justin Becknell
Assistant Professor
Department of Environmental Science
Colby College
(207)-859-5352
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Dr. Pete Countway
Senior Research Scientist
Molecular Lab
Bigelow Lab for Ocean Sci
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